

**A
Synopsis
on
(Local Connect)**

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by

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ABSTRACT

LocalConnect is a web-based local service booking platform designed to connect customers with local service providers such as electricians, plumbers, carpenters, painters, cleaners, and technicians. In traditional methods, customers often depend on personal references, phone calls, or informal contacts to find service providers, which can be time-consuming and may not provide sufficient information about service quality and reliability.

The proposed system provides a centralized platform where customers can search for services, view provider profiles, check ratings and reviews, book services, and track their bookings. Service providers can create profiles, list their services, manage booking requests, and update job status. An administrator can verify service-provider information, manage users and bookings, handle complaints, and monitor the platform.

The system will be developed using modern web technologies such as React.js, Node.js, Express.js, and MongoDB. Authentication and role-based access control will be implemented to improve security. The platform aims to make local service booking easier, faster, more transparent, and convenient for both customers and service providers

INTRODUCTION

3.1 Background

Local service providers play an important role in everyday life. People frequently require services such as electrical repair, plumbing, carpentry, painting, cleaning, appliance repair, and other home maintenance services. However, finding a suitable and reliable service provider is often difficult, especially when an urgent service is required.

At present, many people depend on personal contacts, recommendations from friends or neighbors, advertisements, or telephone calls to find service providers. These methods may require considerable time and do not always provide sufficient information about the provider's experience, availability, pricing, or previous customer feedback.

With the growth of web and mobile technologies, online service platforms can provide a convenient solution to this problem. A digital platform can bring customers and local service providers together in one place and simplify the complete service-booking process.

3.2 Proposed System

LocalConnect is proposed as a web-based platform that connects customers with local service providers. Customers can search for a required service, select a provider, view relevant information, and place a booking request.

The platform will have three major types of users:

Customer

Customers can:

- Register and log in.
- Search for local services.
- View service-provider profiles.
- Check ratings and reviews.
- Book services.
- Track booking status.
- Cancel bookings when applicable.
- Submit ratings and reviews after service completion.

Service Provider

Service providers can:

- Register on the platform.
- Create and manage their profiles.
- Select service categories.
- Add experience and service details.
- Submit required verification information.
- Accept or reject booking requests.
- Update service/job status.
- View customer reviews.

Administrator

The administrator can:

- Manage customers and service providers.
- Review provider verification information.
- Approve or reject providers.
- Manage bookings.
- Handle complaints.
- Monitor platform activities.
- View basic system analytics.

3.3 Service Provider Verification

A major feature of LocalConnect will be a provider verification mechanism. Providers can submit identity and relevant supporting information during registration. Their mobile number and email can be verified, and submitted information can be reviewed by the administrator.

After successful review, the provider can receive a **Verified Provider** status. Customer ratings, reviews, completed bookings, and service history can further help customers evaluate providers.

The system will not claim that online verification guarantees that a provider is completely genuine or risk-free; instead, it will provide multiple trust indicators to help customers make better decisions.

3.4 Basic System Flow

Customer → Search Service → Select Provider → Book Service → Provider Accepts → Service Completed → Customer Review

Provider → Register → Submit Information → Admin Verification → Provider Approval → Receive Bookings

LITERATURE SERVEY

Online service-booking platforms have become an important part of the digital economy. Existing platforms demonstrate how technology can connect customers with professionals and simplify the process of finding and booking services.

Online Service Marketplaces

Existing online service marketplaces provide customers with access to multiple categories of professionals. Users can generally search for a service, compare providers, view ratings, and request or book a service.

These platforms demonstrate the importance of:

- Service categorization
- Provider profiles
- Customer reviews
- Booking management
- Digital communication
- Location-based service discovery

However, large platforms may primarily focus on specific markets, cities, or service categories. A localized platform can focus more specifically on the requirements of local communities.

Service Provider Verification

Trust is an important issue in online service marketplaces. Customers need confidence before allowing an unknown service provider to visit their home or workplace.

Common verification approaches include:

- Mobile and email verification
- Identity-document verification
- Address verification
- Profile verification
- Background checks where available
- Customer ratings and review

LocalConnect incorporates a simplified provider verification workflow suitable for a student project. The administrator can review submitted information before approving a provider.

4.3 Rating and Review Systems

Rating and review systems are commonly used by online platforms to help customers evaluate service quality. A rating system allows customers who have completed a service to provide feedback.

LocalConnect will implement a rating and review mechanism so that customers can evaluate completed services. The information can also help administrators identify poor service quality or repeated complaints.

4.4 Location-Based Services

Location is an important factor when customers need local services. Location-based search can help customers find providers operating in their area and reduce unnecessary travel time.

LocalConnect can use location information such as city, area, and PIN code for service discovery. Map integration can be added as an advanced feature.

4.5 Existing System Limitations

Traditional local service booking commonly has the following limitations:

1. Difficulty finding reliable providers.
2. Lack of centralized provider information.
3. Limited transparency about ratings and reviews.
4. Manual booking through phone calls.
5. Difficulty tracking booking status.
6. Limited mechanism for handling complaints.
7. Lack of structured provider verification.

4.6 Proposed Improvement

LocalConnect aims to address these limitations by combining service discovery, provider profiles, verification, booking management, reviews, and administrative control in a single platform.

The proposed system focuses on providing a simple and localized solution that can later be extended with advanced features such as online payments, real-time location tracking, notifications, chat, and AI-based recommendations.

PROBLEM FORMULATION-NEED AND SIGNIFICANCE

Finding trustworthy local service providers is a common problem faced by households, students, offices, and small businesses. Traditional methods such as personal references and phone calls are not always convenient and may provide limited information about the provider's availability, experience, pricing, or previous performance.

There is a need for a centralized digital platform that allows customers to search for local services, view provider information, make bookings, and track their service requests. At the same time, local service providers need a platform through which they can reach more customers and manage their bookings efficiently.

The significance of LocalConnect is that it provides a structured connection between customers and local service providers. Provider verification, ratings, reviews, booking history, and complaint management can improve transparency and help customers make informed choices.

Therefore, the proposed system aims to provide a convenient, secure, and user-friendly digital solution for local service discovery and booking.

OBJECTIVES

The main objectives of the LocalConnect project are:

1. To develop a web-based platform that connects customers with local service providers.
2. To provide an easy service discovery and booking system where customers can search, compare, and book required services.
3. To implement a provider verification mechanism using contact verification, submitted information, and administrator approval.
4. To provide ratings, reviews, booking history, and status tracking to improve transparency and customer experience.
5. To develop an administrative management system for managing users, providers, bookings, complaints, and basic platform analytics.

METHODOLOGY / PLANNING OF WORK

The LocalConnect project will follow a systematic software development methodology consisting of the following stages.

7.1 Requirement Analysis

The first stage will identify the requirements of customers, service providers, and administrators. Functional and non-functional requirements will be documented in the Software Requirement Specification (SRS).

7.2 System and UI Design

The system architecture, database structure, navigation flow, and user interfaces will be designed before implementation.

Separate interfaces will be designed for:

- Customers
- Service Providers
- Administrators

7.3 Database Design

MongoDB will be used to store application data. Major collections will include:

- Users
- Providers
- Services
- Bookings
- Reviews
- Complaints
- Payments (optional)

Relationships between users, providers, services, bookings, and reviews will be considered during database design.

7.4 Frontend Development

The frontend will be developed using React.js. Responsive interfaces will be created for service search, provider profiles, booking, dashboards, reviews, and administrative functions.

7.5 Backend Development

The backend will be developed using Node.js and Express.js. REST APIs will handle authentication, service management, booking operations, provider verification, reviews, and administrative operations.

7.6 Authentication and Authorization

JWT-based authentication will be used for secure user sessions. Role-based authorization will ensure that customers, service providers, and administrators can access only the functions relevant to their roles.

7.7 Provider Verification

Service providers will submit required information during registration. Mobile/email verification and administrator review will be used to establish a verification status. Approved providers will receive a verified status on the platform.

7.8 Booking and Review Module

Customers will be able to select a provider and submit a booking request. Providers can accept or reject requests and update the service status. After completion, customers can submit ratings and reviews.

7.9 Testing

The system will undergo:

- Unit testing
- API testing
- Integration testing
- Functional testing
- User-interface testing
- Authentication and authorization testing

Postman can be used for API testing.

7.10 Deployment

After successful testing, the application can be deployed using:

- Frontend: Vercel
- Backend: Render
- Database: MongoDB Atlas

The project will be maintained using Git and GitHub for version control.

